1. Problem Statement

Let $\check{\mathcal{U}}_q(\mathfrak{b}_+)$, $\check{\mathcal{U}}_q(\mathfrak{b}_-)$ be the upper and lower Borel subalgebras of the simply-connected quantum group $\check{\mathcal{U}}_q(\mathfrak{sl}(n))$. (This quantum group appears for example in the Klimyk–Schmüdgen book as $U_q^{ext}(\mathfrak{sl}(n))$.)

(1) Construct an explicit algebra isomorphism between $\overline{\mathscr{S}}_n(\mathfrak{B}^*)$ and the generalized quantum double $\check{\mathcal{U}}_q(\mathfrak{b}_-)\bowtie\check{\mathcal{U}}_q(\mathfrak{b}_+)$ (with respect to the standard Drinfeld–Tanisaki skew pairing,

$$\langle \hat{K}_i, \hat{K}_j\rangle = q^{\frac{1}{n}-\delta_{i,j}}, \quad \langle E_i, F_j\rangle = \delta_{i,j}/(q^{-1}-q), \quad \langle \hat{K}_i, F_j\rangle = \langle E_j, \hat{K}_i\rangle = 0,$$

in the notation of Klimyk–Schmüdgen).

(2) Deduce that the algebra $\check{\mathcal{U}}_q(\mathfrak{sl}(n))$ is isomorphic with $\overline{\mathscr{S}}_n(\mathfrak{B}^*)/I$, where I is the two sided ideal generated by $b_{ii}c_{ii}-1$ for $1\le i\le n$, and b_{ij}, c_{kl} are the horizontal arcs directed to the right:

